Name:

Class:

ACTIVITY SHEET

5.3 More practice at calculating quantities

1 Calculate the relative molecular masses of the following molecules, given this data:

$A_r(N) = 14.01$, $A_r(H) = 1.01$, $A_r(S) = 32.06$, $A_r(O) = 16.00$, $A_r(C) = 12.01$.

a ammonia (NH_3)

$M_r(NH_3) =$

$=$

$=$

b sulfuric acid (H_2SO_4)

$M_r(H_2SO_4) =$

$=$

$=$

c carbonic acid (H_2CO_3)

$M_r(H_2CO_3) =$

$=$

$=$

2 Calculate the relative formula masses of the following ionic compounds, given this data:

$A_r(Na) = 22.99$, $A_r(S) = 32.06$, $A_r(O) = 16.00$, $A_r(Mg) = 24.31$, $A_r(N) = 14.01$, $A_r(H) = 1.01$.

a sodium sulfate (Na_2SO_4)

$M_r(Na_2SO_4) =$

$=$

$=$

b magnesium nitrate ($Mg(NO_3)_2$)

$M_r(Mg(NO_3)_2) =$

$=$

$=$

c magnesium hydroxide ($Mg(OH)_2$)

$M_r(Mg(OH)_2) =$

$=$

$=$

3 Calculate the percentage composition of each element in the following compounds, given this data:

$A_r(Mg) = 24.31$, $A_r(Cl) = 35.45$, $A_r(Ca) = 40.08$, $A_r(C) = 12.01$, $A_r(O) = 16.00$, $A_r(Al) = 26.98$, $A_r(S) = 32.06$.

a magnesium chloride ($MgCl_2$)

mass of Mg =

mass of Cl =

formula mass of $MgCl_2$ =

=

% Mg =

% Cl =

b calcium carbonate ($CaCO_3$)

mass of Ca =

mass of C =

mass of O =

formula mass of $CaCO_3$ =

=

% Ca =

% C =

% O =

c aluminium sulfate ($Al_2(SO_4)_3$)

mass of Al =

mass of S =

mass of O =

formula mass of $Al_2(SO_4)_3$ =

=

% Al =

% S =

% O =